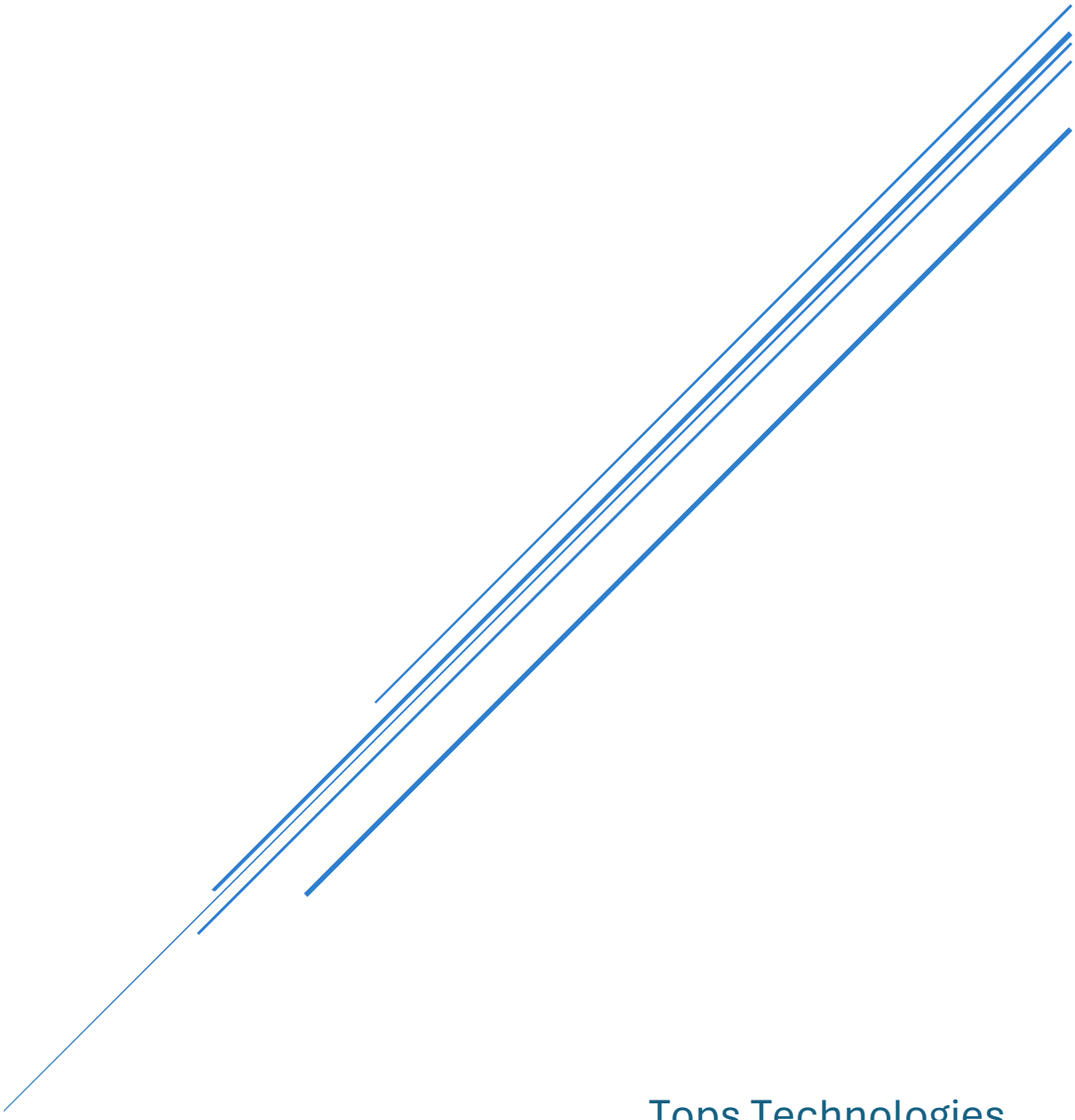
DAY – 3 SWITCH AND ROUTER CONFIGURATION

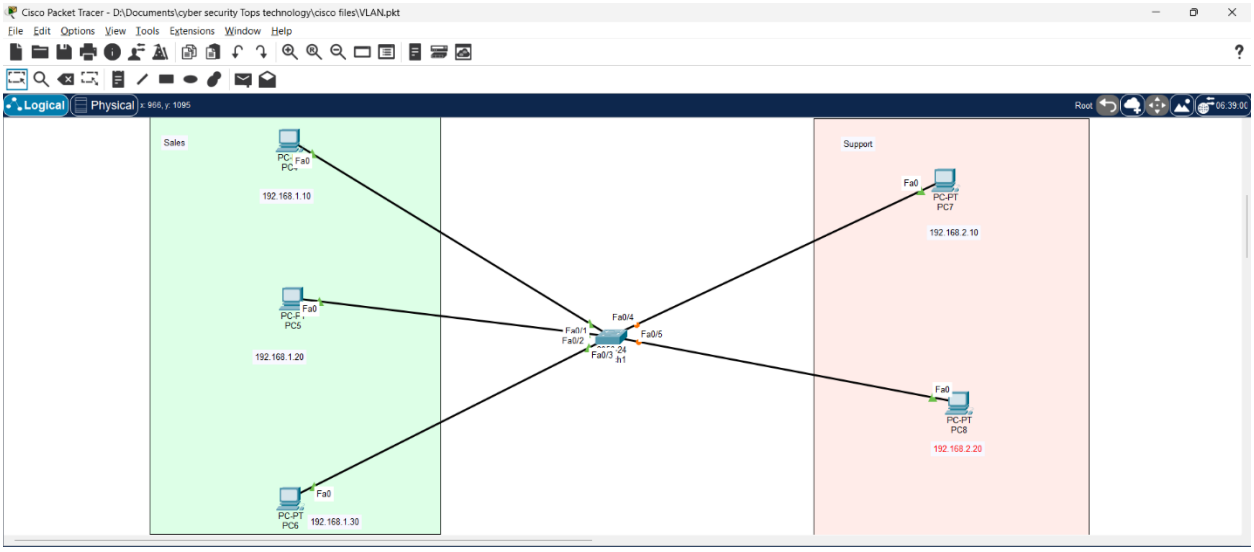
Assignment - 19



Tops Technologies
Cyber Security

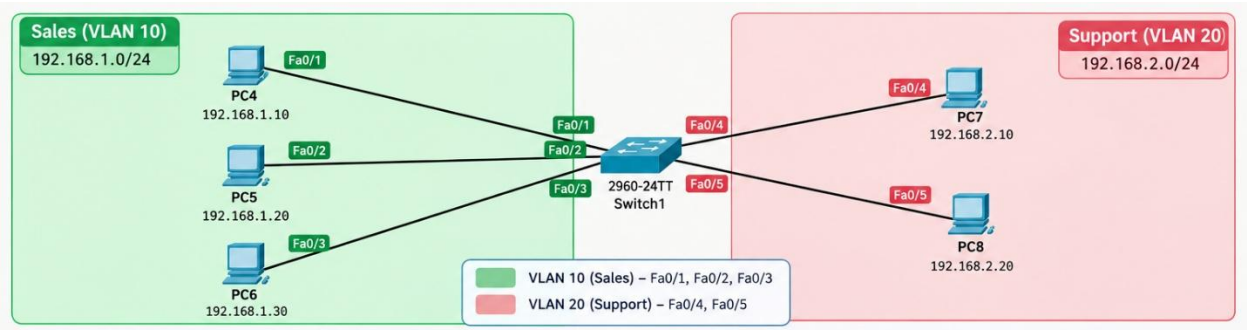
1. **Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.**

➢



2. **Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command. Hint: Use the interface range command to select multiple ports efficiently.**

➢



1. Switch Configuration Commands

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name Support
Switch(config-vlan)# exit

Switch(config)# interface range fastEthernet 0/1-3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit

Switch(config)# interface range fastEthernet 0/4-5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit

Switch(config)# end
Switch#
```

2. Verify VLAN Membership (show vlan brief)

```
Switch# show vlan brief

VLAN  Name                Status   Ports
----- ----------------    -------  ----------------
1     default             active   Fa0/24
10    Sales               active   Fa0/1, Fa0/2, Fa0/3
20    Support             active   Fa0/4, Fa0/5
1002  fddi-default        active
1003  token-ring-default  active
1004  fddinet-default     active
1005  trnet-default       active

Switch#
```

3. Port to VLAN Mapping (Result)

PC	Port	VLAN	IP Address
PC4	Fa0/1	10	192.168.1.10
PC5	Fa0/2	10	192.168.1.20
PC6	Fa0/3	10	192.168.1.30
PC7	Fa0/4	20	192.168.2.10
PC8	Fa0/5	20	192.168.2.20

4. Key Points

- Used interface range to assign multiple ports.
- VLAN 10 (Sales) → Fa0/1-3 (3 ports)
- VLAN 20 (Support) → Fa0/4-5 (2 ports)
- Verified using show vlan brief.

3. **Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.**

➢



ping 192.168.2.10

Reply from 192.168.2.10: bytes=32 time<1ms TTL=127

Reply from 192.168.2.10: bytes=32 time<1ms TTL=127

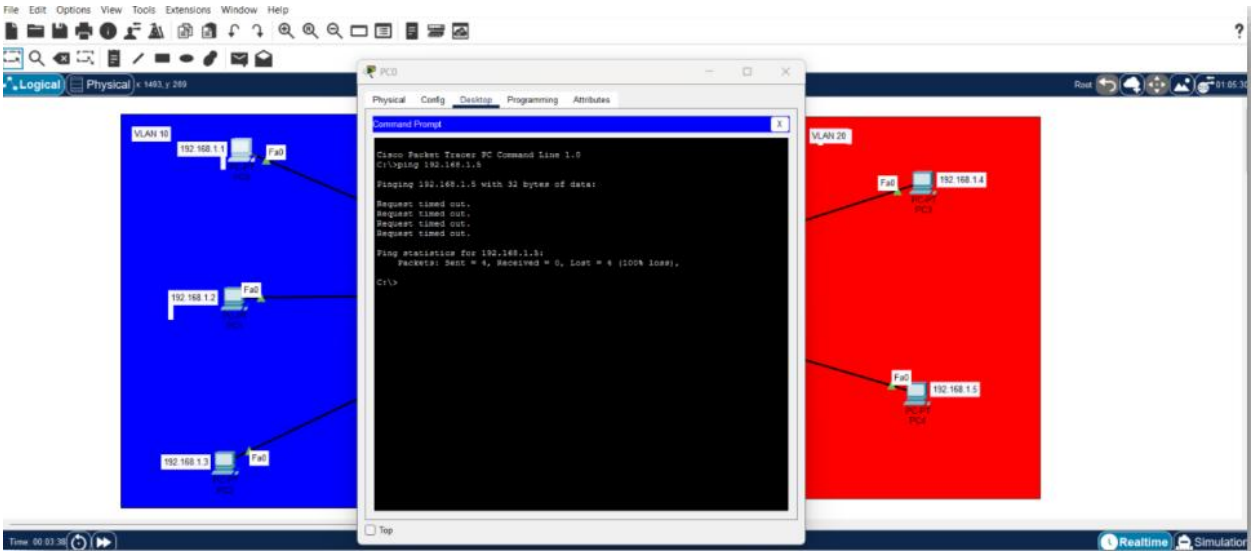
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127

Reply from 192.168.2.10: bytes=32 time<1ms TTL=127

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

4. **Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.

Hint: Think about router-on-a-stick or Layer 3 switch concepts.**



Why does the ping fail?

VLAN 10 (Sales) and VLAN 20 (Support) are separate networks. A Layer 2 switch cannot send data directly between them.

How can we fix it?

We can use Router-on-a-Stick by connecting a router to the switch and configuring sub-interfaces for both VLANs. Alternatively, we can use a Layer 3 switch with routing enabled.

In simple words: Devices in different VLANs cannot communicate directly. We need a router or Layer 3 switch to allow communication between them.